Martin Mbithi Nzilani

Software Engineer

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PROFILE

Software engineer with eight years building web, mobile and cloud systems for government, NGOs and private companies. Currently in the Applications Development section at the Government of Makueni County, part of the in-house team that built myMakueni, the county's enterprise resource planning platform (31 modules, launched August 2026). Strengths in backend and API development, integration with banks and payment gateways, cloud and Linux infrastructure, and secure development. MSc in Cyber Security and Digital Forensics.

CORE COMPETENCIES

Languages: PHP, Python, Java, JavaScript, C, Bash, SQL

Frameworks: Laravel, Django, Flask, Spring Boot, Node.js, React, Vue

Cloud and infrastructure: AWS, Google Cloud, Kubernetes, Docker, CI/CD, Linux and Unix, Nginx

Databases: MySQL, PostgreSQL, MongoDB, Redis

Integration: RESTful APIs, microservices, bank and payment gateway integration, Swagger, Postman

Security and testing: Secure development, system audits, role-based access control, unit and automated testing

Delivery: Agile, Scrum, Kanban, JIRA, Git, GitHub, GitLab, code review

Design: Figma, Adobe XD

PROFESSIONAL EXPERIENCE

Software Engineer, Applications Development

Mar 2023 to present

Government of Makueni County

- Build and maintain the county's in-house software as part of the ICT Applications Development team, including myMakueni, the county ERP launched in August 2026 with 31 modules across finance, planning, health, HR and citizen engagement.
- Engineer the backend services and data models for the public finance modules: revenue billing and receipting, programme-based budgeting and expenditure workflows.
- Design and maintain REST APIs integrating county systems with banks, payment gateways and SMS and email services.
- Deliver the county systems linked from the ERP, including the procurement portal, PMTS, IVAS, Kilimo Makueni and the ECDE, VTC and CIC management systems.
- Optimise SQL and NoSQL schemas and queries to keep reporting responsive as data grows.
- Introduced Git-based workflow, code review and automated testing; mentor junior engineers and work with finance, procurement and operations staff to turn policy into working software.

ICT Officer

Jan 2022 to Feb 2023

ICT Authority, Machakos County Headquarters

- Provided technical support to staff, resolving hardware and software faults.
- Installed and maintained network infrastructure to keep connectivity and system performance stable.
- Ran regular system audits and implemented cybersecurity controls against data breaches and unauthorised access.
- Worked with departmental teams to specify and roll out technology that reduced manual effort.

Mobile Application Developer Jan 2021 to Nov 2021
World Christian Fellowship (NGO)

- Developed and maintained native iOS and Android applications in Swift and Java.
- Implemented interfaces from design mockups and integrated third-party APIs for geolocation and social sharing.
- Tested across devices and OS versions; user satisfaction rose 30% and load times fell 20% after memory profiling.

Web Application Developer Jan 2020 to Jul 2021
Ezana Group

- Developed responsive web applications with HTML, CSS, JavaScript, React, Vue and Bootstrap.
- Worked with UI/UX designers on interfaces that raised user engagement by 20%.
- Built server-side components and REST APIs in Node.js and Express.

Web Developer and ERP Support Officer Jan 2020 to Jan 2021
Ukamba Bible College

- Supported and extended the college's web applications and ERP system.
- Implemented designer mockups as production pages, contributing to a 25% rise in site traffic.
- Reduced page load times, cutting bounce rate by 15%, and resolved cross-browser compatibility issues.

Software Engineer Jan 2018 to Jan 2020
DevLan Solutions Ltd

- Introduced code review, Git version control and unit testing across the team.
- Developed REST APIs for data exchange between software components.
- Profiled and tuned application performance and resolved complex technical faults to reduce downtime.
- Delivered features on Agile cycles aligned to client priorities.

Graphic Designer Jan 2016 to Jan 2018
Lauh Print Company, Machakos

- Produced branding, marketing and digital artwork, and print materials to client brand guidelines, in Photoshop, Illustrator and InDesign.

EDUCATION

MSc, Cyber Security and Digital Forensics, Open University of Kenya

BSc, Computer Science, St Paul's University 2017 to 2020

CERTIFICATIONS

- Software Engineering Certification, Open University of Kenya, 2025
- Agile Project Management and JIRA, HP Foundation, Atlassian and Great Learning, 2024
- DevOps Certification, Edureka, 2022
- Cyber Security Essentials, Cisco, 2019
- Mobile Web Specialist, Andela and Google Africa Developer Scholarship, 2018
- NDG Linux Unhatched and Linux Essentials, Cisco, 2017
- ICDL and Graphic Design, CTC College, 2016